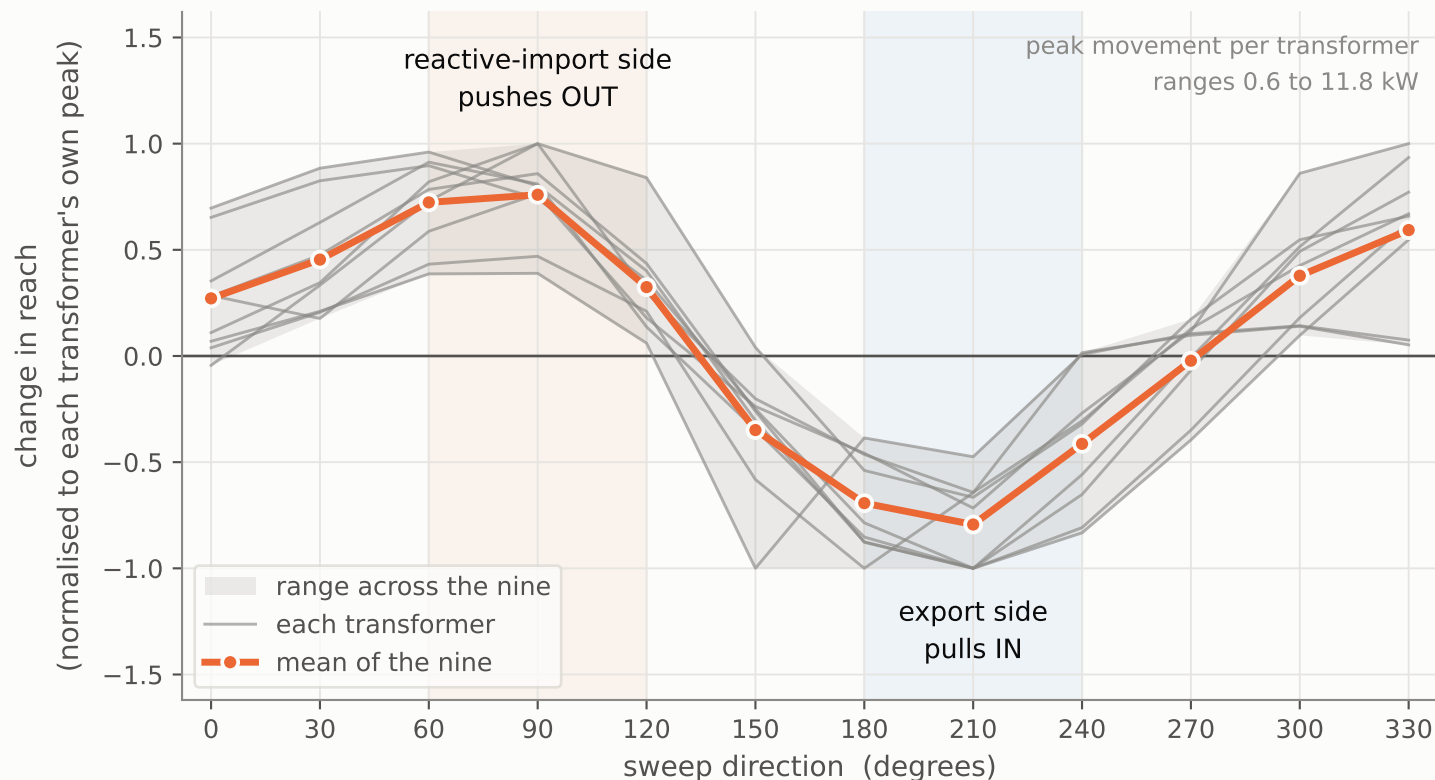
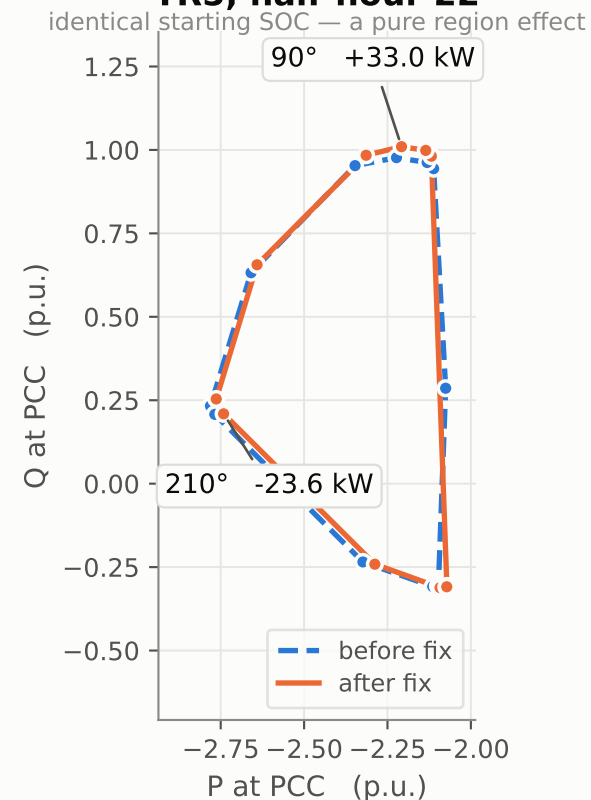


Correcting the load/PV derivatives rotates the flexibility region — consistently, across all nine transformers

Every transformer rotates the same way



TR5, half-hour 22



Left: mean change in the supporting projection per sweep direction, over the half-hours whose committed battery schedule is unchanged. All nine transformers push out on the reactive-import side and pull in on the export side, so the region turns rather than grows. Right: the same effect in the P-Q plane. Sign convention grid → feeder positive, so this region sits in export; 1 p.u. = 1000 kW. Both runs 576/576 Optimal on all nine transformers, exact quadratic formulation, 12 directions at 30°.